

ARKHARA, Russian Federation

WMO: 315940

Lat: 49.417N

Lon: 130.083E

Elev: 135

StdP: 99.72

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -37.0	(c) -34.8	(d) -39.1	(e) 0.1	(f) -36.7	(g) -37.2	(h) 0.1	(i) -34.6	(j) 6.5	(k) -18.0	(l) 5.5	(m) -18.6	(n) 0.4	(o) 320	(p) 0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	29.9	22.4	28.3	21.4	26.9	20.5	23.6	28.3	22.6	27.0	21.6	25.7	2.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	17.1	26.5	21.1	16.0	25.4	20.1	15.0	24.2	71.6	28.4	67.6	27.0	63.9	25.8	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.7	(b) 6.3	(c) 5.3	(e) -39.3	(f) 32.9	(g) 2.9	(h) 1.4	(i) -41.4	(j) 34.0	(k) -43.1	(l) 34.8	(m) -44.7	(n) 35.6	(o) -46.7	(p) 36.7		
(a) 7.7	(b) 6.3	(c) 5.3	(e) -39.4	(f) 26.0	(g) 2.8	(h) 1.4	(i) -41.4	(j) 27.0	(k) -43.1	(l) 27.8	(m) -44.7	(n) 28.7	(o) -46.7	(p) 29.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.4	-24.8	-19.5	-8.8	4.0	12.5	18.4	21.6	19.3	12.5	2.9	-11.1	-23.0		
	DBStd	17.20	4.98	5.75	7.06	4.94	4.20	3.42	2.56	3.16	4.42	5.26	7.38	5.44		
	HDD10.0	4611	1079	827	583	191	22	0	0	1	24	226	634	1023		
	HDD18.3	6752	1338	1061	842	431	188	41	4	26	181	477	884	1282		
	CDD10.0	1113	0	0	0	10	98	252	358	288	99	8	0	0		
	CDD18.3	212	0	0	0	1	6	42	104	55	5	0	0	0		
	CDH23.3	1572	0	0	0	9	88	357	718	366	33	1	0	0		
	CDH26.7	365	0	0	0	1	16	93	185	67	3	0	0	0		
(14) Wind	WSAvg	2.1	1.4	1.8	2.3	2.8	2.7	2.0	1.8	1.7	2.0	2.4	2.1	1.5		
(15) Precipitation	PrecAvg	640	7	7	11	34	53	99	138	141	86	34	18	13		
	PrecMax	840	17	22	37	81	100	188	280	236	218	99	54	30		
	PrecMin	439	0	0	1	0	12	16	35	64	33	1	5	2		
	PrecStd	93	4	5	8	19	23	48	59	47	46	22	13	7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.0	-0.5	10.1	24.6	28.9	32.1	32.4	30.3	26.4	20.6	8.2	-4.1		
		MCWB	-9.2	-3.3	3.9	13.0	16.7	21.7	23.5	23.7	19.7	12.0	3.6	-5.9		
	2%	DB	-11.4	-3.9	6.7	18.4	26.0	29.1	30.3	28.6	24.0	16.5	4.4	-8.9		
		MCWB	-12.3	-5.9	1.9	9.4	15.2	20.5	23.0	22.6	17.6	9.8	1.5	-9.9		
	5%	DB	-14.1	-6.7	4.0	15.3	23.3	27.2	28.8	27.0	22.1	13.8	1.8	-11.6		
		MCWB	-14.9	-8.2	0.2	7.6	13.9	19.6	22.5	21.7	16.5	8.0	-0.5	-12.4		
	10%	DB	-16.3	-9.8	1.8	12.5	20.8	25.2	27.2	25.4	20.0	11.3	-0.4	-14.1		
		MCWB	-16.9	-11.0	-1.3	5.9	12.8	18.8	21.5	20.5	15.2	6.5	-2.0	-14.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.2	-3.1	4.3	13.9	18.0	23.8	25.2	24.9	20.6	13.0	4.2	-5.8		
		MCDB	-8.1	-0.7	9.1	23.8	26.3	29.9	30.2	28.8	25.0	18.7	7.2	-4.4		
	2%	WB	-12.3	-5.9	2.0	10.1	16.2	21.7	23.8	23.4	18.8	10.9	1.6	-9.8		
		MCDB	-11.4	-4.0	6.0	16.8	23.4	27.6	28.8	27.3	22.9	15.6	4.0	-9.0		
	5%	WB	-14.9	-8.2	0.4	8.0	14.8	20.5	23.0	22.2	17.4	8.7	-0.2	-12.3		
		MCDB	-14.2	-6.7	3.6	14.1	21.7	25.8	27.7	26.1	20.9	12.8	1.6	-11.6		
	10%	WB	-16.9	-10.9	-1.1	6.2	13.6	19.3	22.0	21.1	16.0	6.9	-2.2	-14.7		
		MCDB	-16.4	-9.8	1.7	11.9	19.7	24.2	26.2	24.6	18.9	10.9	-0.3	-14.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.0	13.6	12.5	11.4	12.0	10.6	9.4	9.9	12.0	11.2	10.3	11.0		
		MCDBR	12.4	14.0	11.6	15.7	16.1	14.0	12.0	11.9	13.6	14.8	10.7	11.2		
		MCWBR	11.7	12.3	7.8	8.8	8.1	7.0	6.3	6.6	7.8	8.7	8.0	10.5		
	5% WB	MCDBR	12.4	13.7	11.0	14.2	13.8	12.2	10.7	10.4	11.5	12.7	9.5	11.1		
		MCWBR	11.6	12.1	7.5	8.4	7.6	6.7	6.0	6.2	7.2	8.1	7.3	10.4		
(40) Clear-Sky Solar Irradiance	taub		0.283	0.314	0.391	0.537	0.581	0.512	0.490	0.442	0.410	0.424	0.350	0.289		
	taud		2.128	2.037	1.889	1.723	1.714	1.977	2.094	2.219	2.215	2.000	2.069	2.099		
	Ebn at Noon		744	805	791	709	699	760	772	794	777	659	633	666		
	Edh at Noon		81	115	161	213	224	173	152	128	116	120	88	73		
(44) All-Sky Solar Radiation	RadAvg		1.78	3.01	4.48	4.88	5.24	5.63	5.25	4.71	3.80	2.50	1.67	1.38		
	RadStd		0.05	0.07	0.27	0.51	0.46	0.71	0.44	0.34	0.37	0.18	0.12	0.07		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page